

Weekly Meeting Summary: Disturbance-Mode RL-MPC Results

Prepared for supervisor meeting

2026-04-17

Scope

This report is restricted to disturbance-mode results only. Combined-agent, matrix, structured-matrix, and residual notebooks are excluded. The only model-update family kept here is reidentification.

The notebook families covered are:

- distillation: `distillation_RL_assisted_MPC_horizons_unified.ipynb`, `distillation_RL_assisted_MPC_distillation_RL_assisted_MPC_weights_unified.ipynb`, `distillation_RL_assisted_MPC_reidentification.ipynb`
- polymer: `RL_assisted_MPC_horizons_unified.ipynb`, `RL_assisted_MPC_horizons_dueling_unified.ipynb`, `RL_assisted_MPC_weights_unified.ipynb`, `RL_assisted_MPC_reidentification_unified.ipynb`

For each included method, the figures focus on:

- average reward comparison against the paired MPC bundle
- last-episode MPC comparison

Selection Notes

- Distillation is presented first, per your request.
- The latest stored April 16, 2026 distillation horizon, dueling-horizon, and weights runs use the newer unified reward profile introduced on April 16, 2026.
- The latest stored disturbance reidentification result for distillation is still the TD3 run from April 16, 2026.
- On polymer, I found qualifying disturbance bundles for DQN horizon, dueling horizon, weights, and reidentification.

Executive Summary

- Distillation horizon adaptation is currently the strongest family in the report. The April 16, 2026 DQN and dueling runs both beat the paired MPC baseline, and the dueling variant is more concentrated around a small set of horizon recipes.

- Distillation SAC weights are also positive on the newer reward profile, with final gap +1.520 and last-50 gap +2.272.
- Distillation reidentification is still the most mixed result: strong mid-run collapse followed by recovery to a positive final gap +0.060.
- Polymer dueling horizon is still the best polymer disturbance result on disk. It beats the paired MPC baseline in 190 of 200 episodes and ends with final gap +1.692.
- Polymer weights are also clearly positive, with final gap +1.515 and last-50 gap +1.732.
- Polymer DQN horizon remains positive and clearly beats MPC, but it is now weaker than the latest polymer dueling and weights runs.
- Polymer reidentification is still slightly negative at the end, with final gap -0.070 , so polymer remains a weak benchmark for evaluating the model-update idea.

Distillation

Method	Agent	RL Final	MPC Final	Final Gap	Better Eps
DQN horizon	DDQN	19.433	16.079	+3.354	118/200
Dueling horizon	Dueling DDQN	18.914	16.079	+2.835	186/200
Weights	SAC	17.599	16.079	+1.520	169/200
Reidentification	TD3	-0.205	-0.265	+0.060	70/200

Table 1: Latest stored distillation disturbance runs with paired MPC comparison.

DQN Horizon

The latest stored disturbance DQN horizon run is `distillation_horizon_disturb_fluctuation_standard_unifi` compared against `distillation_compare_horizon_disturb_fluctuation_standard/20260416_192443`.

- Final reward is 19.433 versus paired MPC 16.079, so the final gap is +3.354. The last-50 gap is still positive at +0.299.
- The nominal MPC pair (6, 3) is the most used recipe, but only at 7.88% of logged selections. The policy remains broad rather than collapsing to one horizon pair.
- The best reward occurs at episode 131, but the run still has large negative excursions, with worst gap -11.253 at episode 51.

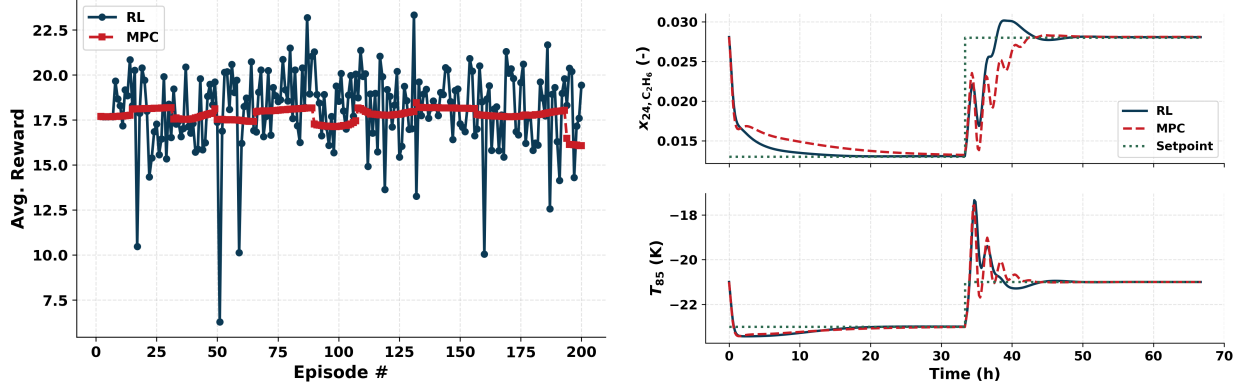


Figure 1: Distillation DQN horizon: average reward comparison, last-episode MPC comparison.

Dueling Horizon

The latest stored disturbance dueling-horizon run is `distillation_dueling_horizon_disturb_fluctuation_standard/20260416_192310` compared against `distillation_compare_dueling_horizon_disturb_fluctuation_standard/20260416_192310`.

- Final reward is 18.914 versus paired MPC 16.079, for final gap +2.835. The last-50 gap is much stronger at +2.720, and RL is better in 186 of 200 episodes.
- The policy is much more concentrated than plain DQN horizon: (6, 3) accounts for 53.36% of logged selections, followed by (8, 2) at 20.27% and (13, 5) at 16.12%.
- This currently looks like the more stable distillation horizon method: smaller final gap than DQN, but a much stronger late-run gap.

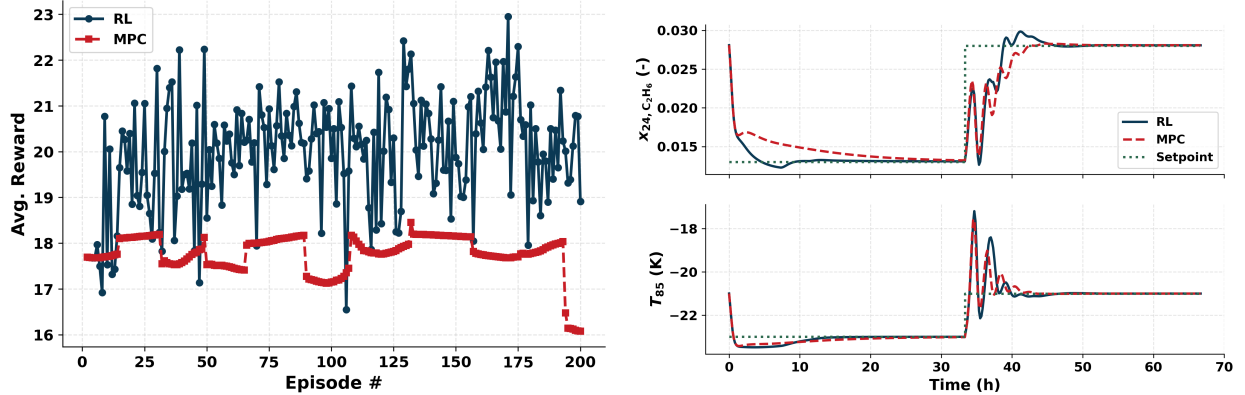


Figure 2: Distillation dueling horizon: average reward comparison, last-episode MPC comparison.

Weights

The latest stored disturbance weights run is the SAC result `distillation_weights_sac_disturb_fluctuation_standard/20260416_192602` compared against `distillation_compare_weights_sac_disturb_fluctuation_standard/20260416_192602`.

- Final reward is 17.599 versus paired MPC 16.079, so the final gap is +1.520. The last-50 gap is larger at +2.272, which means the late part of training is stronger than the final episode alone suggests.
- The four logged penalty multipliers average [1.146, 1.162, 0.999, 1.033] and end at [0.837, 1.100, 0.761, 1.014].
- This is a useful positive result on the new reward profile even though the latest qualifying run is SAC rather than TD3.

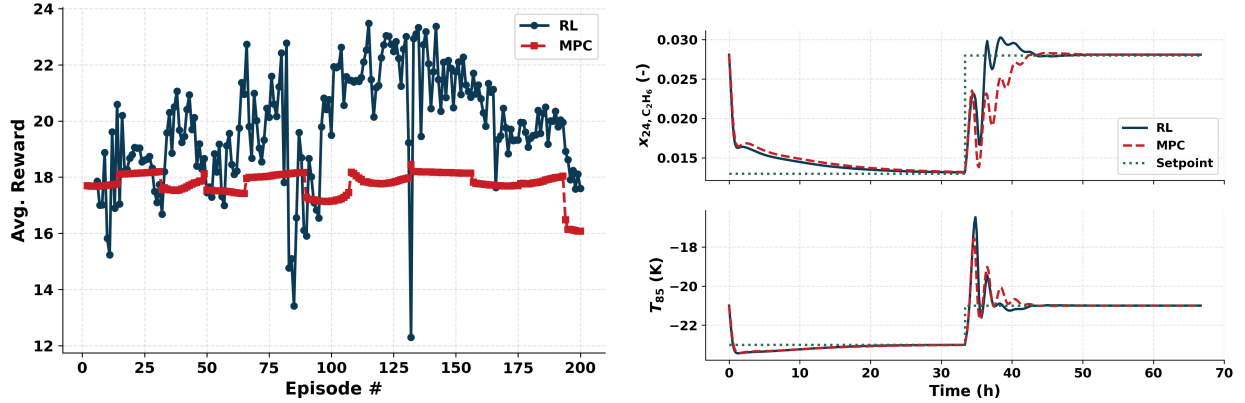


Figure 3: Distillation weights: average reward comparison, last-episode MPC comparison.

Reidentification

The latest stored disturbance reidentification run is `distillation_reidentification_td3_disturb_fluctuation`, compared against `distillation_compare_reidentification_td3_disturb_fluctuation_mismatch/20260416_1`.

- Final reward is -0.205 versus paired MPC -0.265 , for final gap $+0.060$. The last-50 gap is $+0.025$.
- The main issue is still the mid-run collapse. The worst gap is -4.237 at episode 51, after which the run recovers and finally reaches its best reward at episode 200.
- The blend factors stay moderate overall: $\eta_A \approx 0.323$ mean and 0.385 last, $\eta_B \approx 0.355$ mean and 0.398 last. Mean coefficient clipping is only 2.58%, much lower than the polymer reidentification case.

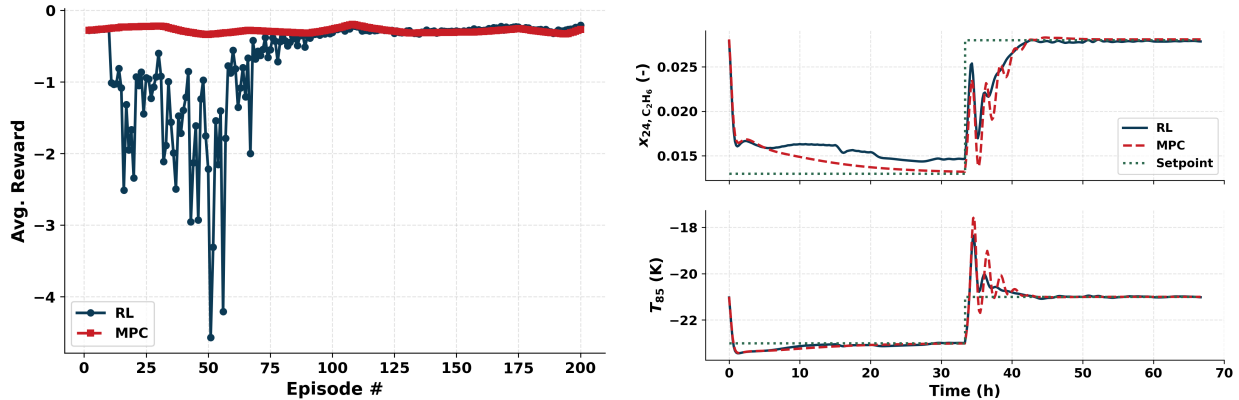


Figure 4: Distillation reidentification: average reward comparison, last-episode MPC comparison.

Polymer

Method	Agent	RL Final	MPC Final	Final Gap	Better Eps
DQN horizon	DDQN	-2.533	-3.843	+1.310	196/200
Dueling horizon	Dueling DDQN	-2.726	-4.417	+1.692	190/200
Weights	TD3	-2.903	-4.417	+1.515	195/200
Reidentification	TD3	-4.487	-4.417	-0.070	10/200

Table 2: Latest stored polymer disturbance runs with paired MPC comparison.

DQN Horizon

This notebook changes the MPC prediction and control horizons online with a DQN policy. The latest stored disturbance run is `horizon_disturb_unified/20260407_175020`, compared against `disturb_compare_horizon_unified/20260407_175031`.

- Final reward is -2.533 versus paired MPC -3.843 , so the final gap is $+1.310$. The last-50 gap is $+1.213$, and RL is better in 196 of 200 episodes.
- The nominal MPC pair (9, 3) is still the most used recipe, but only at 3.22% of logged horizon selections. The policy is spread broadly over many longer-horizon combinations.
- The best reward occurs late at episode 197. This remains a strong polymer disturbance result, but it is now behind the latest dueling and weights runs.

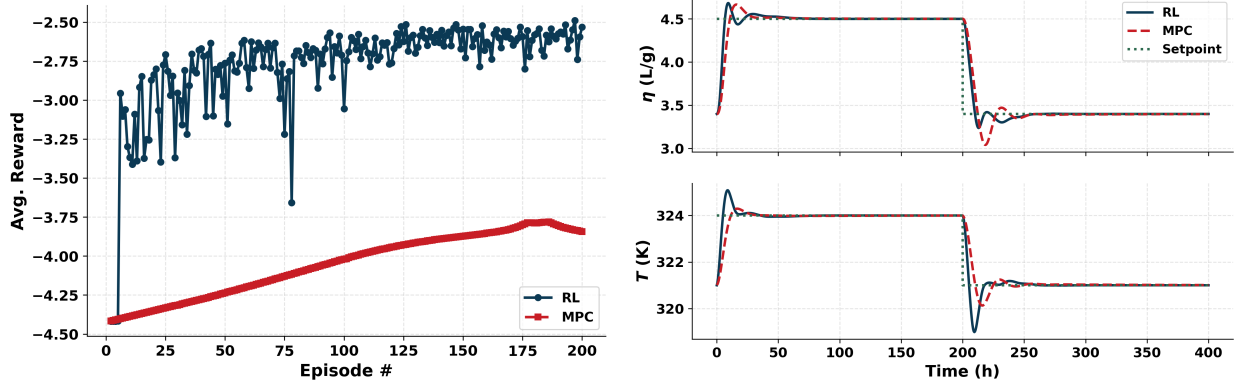


Figure 5: Polymer DQN horizon: average reward comparison, last-episode MPC comparison.

Dueling Horizon

This notebook uses the dueling DDQN horizon supervisor. The latest stored disturbance run is `dueling_horizon_disturb_unified/20260417_035217`, compared against `disturb_compare_dueling_horizon_u`

- Final reward is -2.726 versus paired MPC -4.417 , so the final gap is $+1.692$. The last-50 gap is $+1.745$, and RL is better in 190 of 200 episodes.
- Unlike the distillation dueling case, polymer does not concentrate on the nominal MPC pair (9, 3). The nominal pair is only 5.79% of logged horizon selections and 0.83% of decision points.
- The most-used horizon recipes are all non-nominal, led by (19, 5), (9, 8), and (17, 4). This is still the strongest polymer disturbance result currently stored.

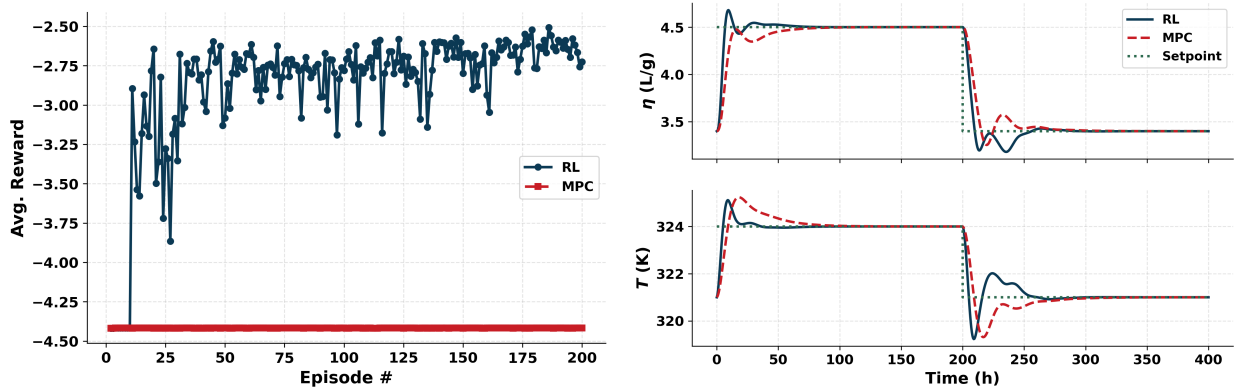


Figure 6: Polymer dueling horizon: average reward comparison, last-episode MPC comparison.

Weights

This notebook lets TD3 scale the output and input penalty weights online. The latest stored disturbance run is `td3_weights_disturb/20260417_031734`, compared against `disturb_compare_td3_weights/2026041`

- Final reward is -2.903 versus paired MPC -4.417 , so the final gap is $+1.515$. The last-50 gap is $+1.732$, and RL is better in 195 of 200 episodes.
- The four logged penalty multipliers average $[1.403, 1.495, 1.263, 1.333]$ and end at $[1.957, 1.703, 1.176, 1.246]$.
- This is a clear positive polymer result, but it still trails the latest dueling-horizon run on final reward gap.

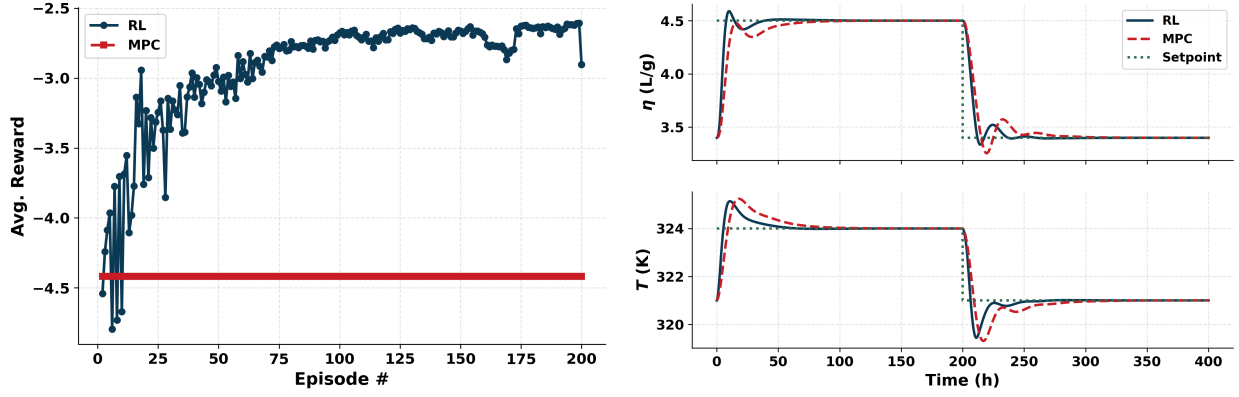


Figure 7: Polymer weights: average reward comparison, last-episode MPC comparison.

Reidentification

This notebook identifies low-rank model corrections online and lets TD3 blend the nominal and identified models. The latest stored disturbance run is `td3_reidentification_disturb/20260415_120803`, compared against `disturb_compare_td3_reidentification/20260415_120810`.

- Final reward is -4.487 versus paired MPC -4.417 , so the final gap is -0.070 . The last-50 gap is -0.087 , and RL is better in only 10 of 200 episodes.
- The mean blend factors are $\eta_A \approx 0.421$ and $\eta_B \approx 0.411$, but by the end the policy shifts strongly toward B : $\eta_A = 0.108$, $\eta_B = 0.846$.
- Mean coefficient clipping fraction is 12.18%, much higher than the distillation reidentification run. Polymer is still not giving a useful positive signal for this family.

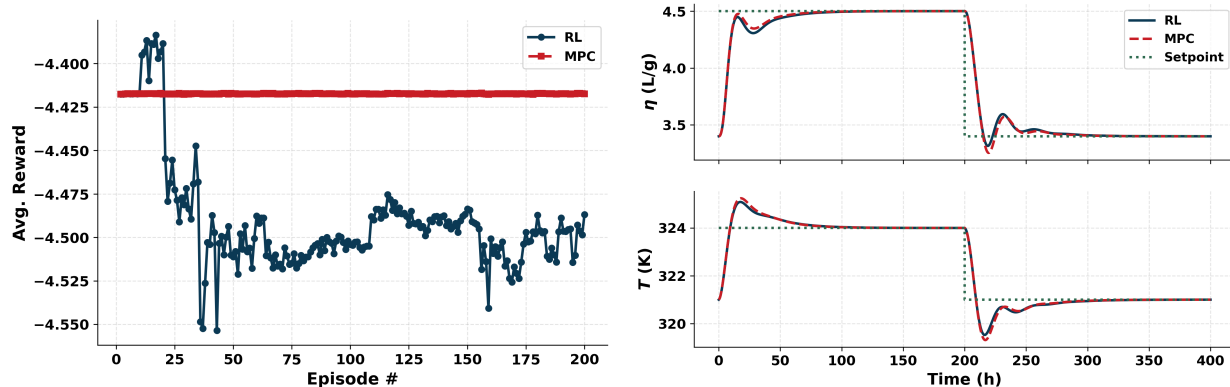


Figure 8: Polymer reidentification: average reward comparison, last-episode MPC comparison.

Van de Vusse Motivation And Current Stage

The current motivation for Van de Vusse is still the same: polymer is giving a good signal for horizon and weight adaptation, but it is still a weak benchmark for the model-update idea. Polymer reidentification remains slightly worse than MPC, and the removed matrix-style families were also not giving strong gains. That makes it hard to judge whether the adaptation idea itself is weak or whether the polymer benchmark is simply not sensitive enough.

Van de Vusse is therefore being developed as a benchmark where operating-point changes should create a clearer need for model adaptation. The current stage is still pre-RL:

- nonlinear plant added under `systems/vandevusse/`
- direct-local-linearization system identification in place
- corrected benchmark feed-temperature handling
- offset-free MPC baseline notebook running in disturbance mode with the canonical `ca0_blocks` disturbance
- observer-update cleanup and observer-pole search utilities added
- no Van de Vusse RL result bundle yet, so it is not part of the tables